Cointegration-Based Strategies in Forex Pairs Trading

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Abstract

Pairs trading exploits pricing differentials between related assets. A cointegration-based pairs strategy selects pairs whose residual spread looks mean-reverting enough to trade, so that short-term divergences are more plausibly exploitable. This paper studies whether that filter improves risk-adjusted performance in the foreign-exchange market. Using daily spot prices for the seven most liquid USD crosses over 2007–2025, we compare a simple always-trade pairs rule with an Engle–Granger–style **tradability filter** on rolling 257/21-day train/test windows, across z-score entry thresholds ± 1, ± 2, and ± 3. Charging a stylized 2 bp round-trip cost on position changes, the unlevered cointegration portfolio earns about 0.5% annualized at ± 1 (and 0.2–0.3% at wider bands) because capital is rarely deployed: invested fraction is about 8% of pair-days at ± 1. It still delivers higher Sharpe and Sortino ratios than the simple benchmark at every threshold. A paired block-bootstrap test of the portfolio Sharpe and Sortino differences finds a statistically significant edge at the pre-committed headline threshold ± 1; at ± 2 and ± 3 the ranking is consistent in point estimates but not statistically separable. The ranking is unchanged relative to frictionless (zero-cost) results and survives a grid up to 5 bp. A companion that sizes each live pair to 3% train-window volatility still ranks cointegration first on Sharpe, with about 4% annualized return at about 8% book volatility.

Keywords: Cointegration, Pairs Trading, Mean Reversion, Forex, Currency Markets
JEL classification: C58, F31, G11, G15

1 Introduction

Traders continually search for relative-value inefficiencies. Pairs trading is a leading example: identify assets that usually move together, and trade temporary divergences in anticipation of realignment (Gatev et al., 2006; Krauss, 2017).

Conventional implementations lean on correlation or distance statistics and treat mean reversion as a short-horizon regularity. Cointegration reframes the problem. If two non-stationary price series share a stationary linear combination, deviations from that equilibrium are statistically constrained and therefore more credible as trading signals (Engle & Granger, 1987; Vidyamurthy, 2004).

Despite a large equity literature, cointegration-based pairs trading remains comparatively thin in FX. This paper fills that gap for major currencies. The seven most liquid USD crosses are natural candidates for relative-value analysis: they share a common dollar numeraire, are tightly linked by trade and capital flows, and are disciplined by parity relations such as purchasing-power parity and interest-rate parity (Engel & West, 2005; Fama, 1984; Rogoff, 1996). Those links make long-run co-

movement among majors **plausible**; they do not, by themselves, deliver a trading rule. We therefore keep the empirical design statistical: we compare (i) a **simple** pairs strategy that always trades the synthetic spread on every out-of-sample block with (ii) a **cointegration-based** strategy that trades only when an Engle–Granger–style tradability filter passes on the preceding training window. The universe comprises all unordered pairs among seven USD-denominated majors ($C(7,2) = 21$ pairs). Performance is evaluated with annualized return and volatility, Sharpe, Sortino, and Calmar ratios, and maximum drawdown (Sharpe, 1966; Sortino & Meer, 1991; Young, 1991) on **unlevered** portfolios, together with invested-fraction and active-day occupancy. Cumulative-return figures scale the cointegration path to equal ex-post volatility for visual comparison only.

Portfolio metrics and Figure 3–Figure 5 below are produced by the companion replication code on a frozen Yahoo Finance sample.

2 Literature review

(Krauss, 2017) organizes pairs-trading research into distance, time-series, stochastic-control, cointegration, and other approaches.

The distance method of (Gatev et al., 2006) forms pairs by historical Euclidean proximity and opens when spreads widen. It is simple and relatively robust to data-snooping, but the static L^2 metric is outlier-sensitive and poorly adapted to regime shifts.

(Elliott et al., 2005) model the spread as a mean-reverting Gaussian Markov chain in state space. The framework is tractable, yet Gaussian assumptions can fail in FX, especially in stress episodes.

(Jurek & Yang, 2007) study stochastic control for arbitrageurs choosing between a mean-reverting spread and a risk-free asset, obtaining closed-form policies under Ornstein–Uhlenbeck uncertainty.

(Vidyamurthy, 2004) provides the cointegration blueprint: pair preselection, Engle–Granger tradability filtering, and nonparametric entry/exit rules. (Rad et al., 2015) implement distance preselection plus Engle–Granger and find return profiles similar to pure distance methods. (Galenko et al., 2012) use multivariate cointegration weights on ETF baskets and document mean-reverting portfolio returns, with the usual caveat that performance can weaken outside the estimation regime.

Related applications include equity hedging via cointegration (Burgess, 2003), gold–silver parity (Liu & Chou, 2003), inflation hedges (Bampinas & Panagiotidis, 2015; 2016), and currency-portfolio optimization (Dunis et al., 2011). (Huck & Afawubo, 2015) compare distance, stationarity, and cointegration on S&P 500 constituents and conclude that cointegration delivers more stable excess returns after costs; our contribution is the analogous horse race in major FX. Ranking/forecasting approaches (Huck, 2009; 2010) are complementary but outside our design.

Relative to short-horizon correlation filters, cointegration anchors trades in a long-run equilibrium. In FX that equilibrium has a natural economic reading via parity and common macro drivers; we spell out that motivation—and what we do **not** estimate—in the next section.

3 Methodology

3.1 Data

We use daily adjusted closes for EUR/USD, GBP/USD, USD/JPY, USD/CHF, USD/CAD, AUD/USD, and NZD/USD from Yahoo Finance (Yahoo Finance, 2025), January 1, 2007 to December 31, 2025. Quotes with USD as numerator are inverted so every series is XXXUSD

(e.g.
JPYUSD $= \frac{1}{\text{USDJPY}}$) (Pound Sterling Live, 2022). Analysis uses log prices.

Prices are processed in rolling blocks: a training window of $n = 257$ days (about one year) followed by a testing window of $m = 21$ days (about one month). After each test block, windows advance by m days so the next train ends where the previous test ended (Figure 1). The training sample is used to (i) run the tradability filter when required and (ii) estimate the spread mean and standard deviation used in the out-of-sample z-score. The simple strategy skips (i) but still uses the training moments.

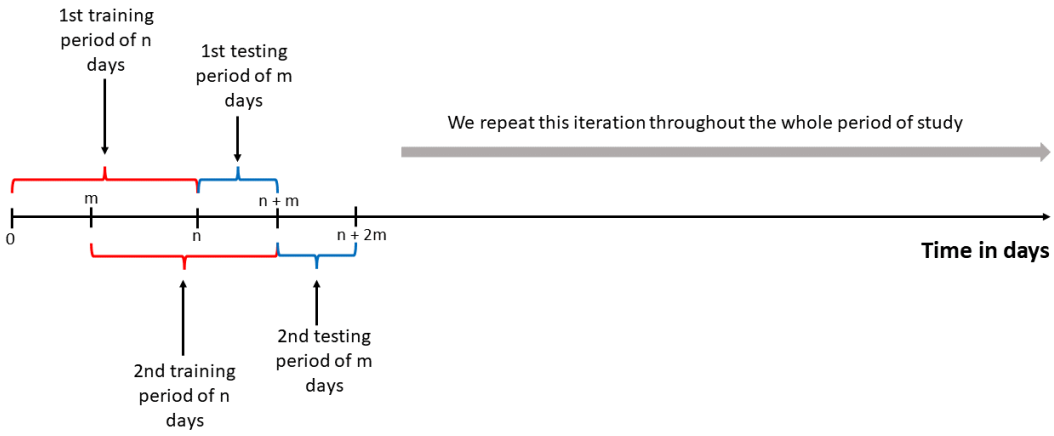


Figure 1: Iterative training and testing windows over the sample.

3.2 Economic motivation

Why should an Engle–Granger filter be more than a statistical device among 21 correlated USD crosses? Purchasing-power parity (PPP) links national price levels to exchange rates; although real exchange rates mean-revert only slowly, the associated "PPP puzzle" still implies that related nominal rates are not free to wander independently forever (Rogoff, 1996). Covered and uncovered interest-rate parity connect spot rates, forwards, and interest differentials; even when uncovered interest parity fails as a short-horizon forecast (Fama, 1984), exchange rates remain asset prices tightly tied to macro fundamentals (Engel & West, 2005). Together with a shared USD numeraire and deep G10 liquidity, these parity and macro links make **long-run co-movement**—and therefore cointegration-based relative-value trading—economically plausible in this universe (Dunis et al., 2011).

Importantly, we do **not** estimate PPP baskets, trade real-exchange-rate gaps, or form positions from UIP residuals. The trading object remains a bivariate log-price spread among unordered XXXUSD majors, gated by the Engle–Granger filter described below. Parity relations motivate **where** we look; the pre-committed EG rule determines **when** we trade. At this stage, parity-based filters are left as a natural extension.

3.3 Cointegration analysis

We gate trading with the Engle–Granger two-step procedure (Engle & Granger, 1987), used here as a **tradability filter** in the sense of (Vidyamurthy, 2004) rather than as a formal size-correct cointegration test. Johansen methods can recover multiple relations in larger systems; for bivariate FX pairs we prefer the Engle–Granger blueprint for transparency and leave Johansen to future work.

Step 1. Augmented Dickey–Fuller (ADF) tests on each log-price series. Failure to reject the unit-root null supports treating both series as $I(1)$.

Step 2. OLS long-run regression

$$y_t = \alpha + \beta x_t + \varepsilon_t \tag{1}$$

with residual

$$\hat{\varepsilon}_t = y_t - \hat{\beta} x_t - \hat{\alpha} \tag{2}$$

.

Step 3. ADF test on $\hat{\varepsilon}_t$. We pass the window when the residual ADF p-value is at most 0.05 and use $\hat{\beta}$ as the hedge ratio for the synthetic spread. Because β is estimated rather than known, that residual statistic does **not** follow the Dickey–Fuller distribution of Step 1: a size-correct Engle–Granger test would compare it against MacKinnon cointegration critical values, which are stricter ((MacKinnon, 1994; 2010); about -3.36 versus -2.87 at the conventional 5% level for $N = 2$ with a constant on our 257-day windows). We deliberately keep the looser ordinary-ADF residual cutoff as a **pre-committed tuning parameter** that controls how permissive the filter is. We do not claim that a pass means "cointegration detected at 5%"; a pass means the residual looks stationary enough, under this operational rule, to trade the spread out of sample.

Cointegration is a **relation**, not a directed trading object, so we work with $C(7, 2) = 21$ unordered pairs. Legs are ordered alphabetically by ticker so that each pair has a unique synthetic spread $\log P^1 - \beta \log P^2$. The residual filter can depend on which series is the regressand; as a fixed design choice we therefore run **both** OLS orientations on each training window and, among those that clear the residual cutoff, retain the orientation with the clearer residual ADF statistic (more negative t -stat), mapping that hedge into the alphabetical spread. If neither orientation passes, the pair is flat for the following test window.

3.4 Mean-reversion strategy

On each testing day the z-score of the spread is

$$Z_t = \frac{\hat{\varepsilon}_t - \mu}{\sigma} \tag{3}$$

where μ and σ are the training mean and standard deviation of $\hat{\varepsilon}_t = \log P_t^1 - \beta \log P_t^2$ (intercept cancels in the z-score).

Enter a long spread when $Z_t < -z^\star$ and a short spread when $Z_t > z^\star$, for thresholds $z^\star \in \{1, 2, 3\}$. Signals are lagged two days to allow for execution delay:

$$R_t^{\text{gross}} = \text{Signal}_{t-2} \times (r_t^1 - r_t^2) \tag{4}$$

with equal notional on each leg (not β-hedged PnL). Flat days enter the Sharpe denominator as zeros.

Transaction costs. Let $s_t \in \{-1, 0, 1\}$ denote the lagged signal. We charge a round-trip cost of κ basis points of pair notional on position changes,

$$\text{cost}_t = (\kappa / 10^4) \times |s_t - s_{t-1}| \ / 2 \tag{5}$$

, so that opening or closing ($|\Delta s|$=1) costs $\kappa/2$ bp and a long–short flip ($|\Delta s|$=2) costs a full κ bp. Net returns are

$$R_t^{\text{net}} = R_t^{\text{gross}} - \text{cost}_t \tag{6}$$

. We report the grid $\kappa \in \{0, 1, 2, 5\}$, with **headline** results at $\kappa = 2$ bp (a stylized all-in estimate for liquid G10 spot, not a full bid–ask plus rollover model). Zero cost recovers the frictionless horse race; 5 bp is a conservative upper bound. Swap/rollover is left as a caveat (short test windows and frequent flats keep overnight exposure limited).

3.5 Strategy implementation and portfolio construction

Simple pairs. Trade every testing window for every unordered pair, using training moments only for the z-score (alphabetical OLS hedge, no cointegration gate).

Cointegration-based pairs. Trade a testing window only if the tradability filter above passes on the preceding training window; otherwise the pair is flat.

Pair returns are summed and divided by 21 to form a standardized portfolio. Main tables report these series **unlevered** and, unless noted, **after** $\kappa = 2$ bp costs. Occupancy is the mean **invested fraction**: the average across days of (number of pairs with a nonzero lagged signal) /21. **Active days** are the share of calendar days on which at least one pair is live. Flat days enter the Sharpe denominator as zeros, so unlevered volatility is occupancy-diluted. As an implementable **companion**, on each 257-day training block we estimate pair spread volatility $\hat{\sigma} = \mathrm{std}(r_1 - r_2)\sqrt{252}$ and, on the following 21-day test block, size every live pair by $L = 0.03/\hat{\sigma}$—the same L for cointegration and simple (mean $L \approx 0.33$). Pair PnL is L times the net equal-notional return (costs $\kappa = 2$ bp scale with L); flat days stay at 0. The companion book is the **sum** of the 21 pair PnLs, not 1/21. The 3% per-position cap is an arbitrary choice. We do not force the two books to the same occupancy, so book volatility follows how many pairs are live (Table 10). Cumulative-return figures separately rescale the cointegration path so that **daily** volatilities match the simple path (Table 1); those figures are visual aids only.

z-score	**Equal-vol scale L**	**Ann. vol EG (%)**	**Ann. vol simple (%)**
± 1	4.50	0.89	4.02
± 2	4.75	0.60	2.85
± 3	5.39	0.38	2.06

Table 1: Equal ex-post daily-vol scales used in cumulative-return figures only ($\kappa = 2$ bp; replication sample).

The 21 unordered pairs are not independent experiments. Every leg is a USD cross, and many pairs share a second currency, so pair returns comove under common dollar and risk shocks. Pair-level Sharpe ratios are therefore descriptive of the cross-section, not twenty-one separate hypothesis tests. We do not apply multiple-testing adjustments or clustered inference to the pair panel. Our primary evidence is the **portfolio** comparison of Engle–Granger versus always-trade rules (equal weight 1/21), which already aggregates those dependent legs. On those two daily portfolio series we test whether the Sharpe and Sortino differences are distinguishable from zero with a paired circular block bootstrap that resamples calendar dates once and applies the same blocks to both books, preserving contemporaneous cross-book correlation (Ledoit & Wolf, 2008). Block length equals the 21-day test window; we studentize the Sharpe difference with a Newey–West HAC standard error (Newey & West, 1987) (Bartlett kernel; lag chosen by the standard $4(T/100)^{2/9}$ rule) and report a percentile interval for the Sortino difference (no closed-form HAC under our downside definition: annualized return over the standard deviation of strictly negative daily returns). Calmar ratios remain descriptive: maximum drawdown is path-dependent, and block reshuffling would scramble that path. ± 1 is the pre-committed headline design; ± 2 and ± 3 are reported alongside it (Table 6).

4 Results

4.1 Descriptive analysis

Per-currency return paths and additional descriptive heatmaps are omitted for brevity. Figure 2 shows pair-level annualized Sharpe ratios at the baseline threshold $z^\star = 1$ after $\kappa = 2$ bp costs.

Dispersion across unordered pairs is large; a few crosses (notably some GBP and JPY combinations in this sample) dominate the right tail of risk-adjusted pair performance.

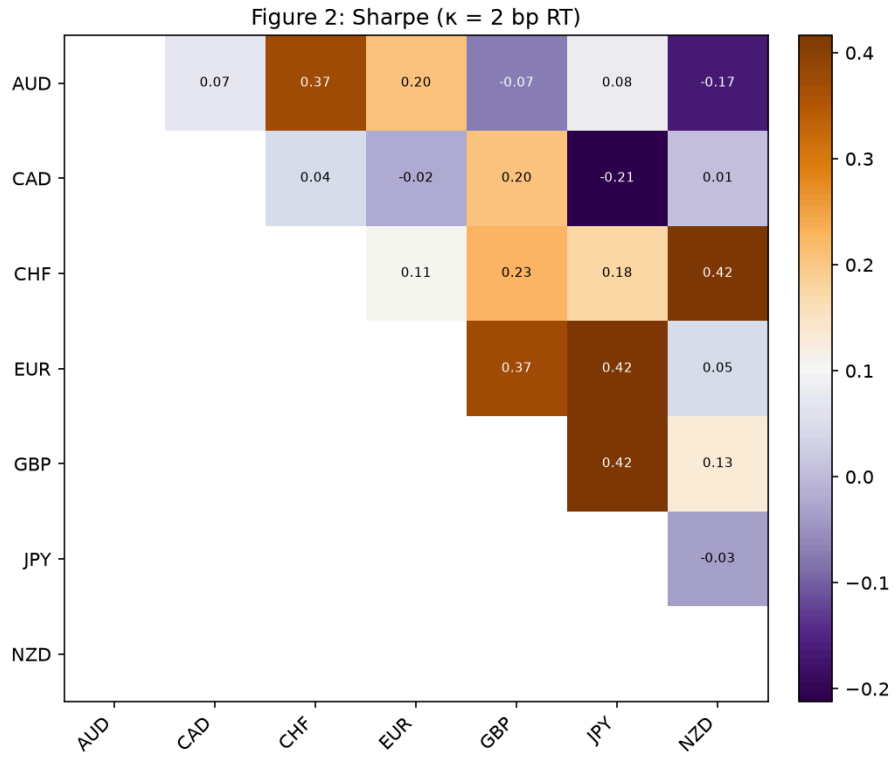


Figure 2: Heatmap of annualized Sharpe ratios by unordered pair ($z^\star = 1$, EG-filtered, $\kappa = 2$ bp RT).

4.2 Cointegration-based strategy versus simple pairs

We compare **unlevered** portfolios at training/testing windows 257/21. Headline tables use $\kappa = 2$ bp; Table 8 reports Sharpe across the full cost grid. Table 2 and Figure 3 give the $z^\star = 1$ results (figures use equal ex-post vol scaling on net returns).

Metric	Cointegration	Simple
Annualized return (%)	0.49	−0.17
Annualized volatility (%)	0.89	4.02
Sharpe ratio	0.55	−0.04
Sortino ratio	0.63	−0.05
Calmar ratio	0.20	−0.01
Maximum drawdown (%)	−2.48	−15.89
Invested fraction (%)	8.24	51.27
Active days (%)	68.33	94.58

Table 2: Unlevered performance at $z^\star = \pm 1$ after $\kappa = 2$ bp RT costs (windows 257/21, 21 unordered pairs, sample through 2025). Occupancy from lagged signals (independent of κ). Frictionless Sharpes are 0.62 (EG) and 0.01 (simple).

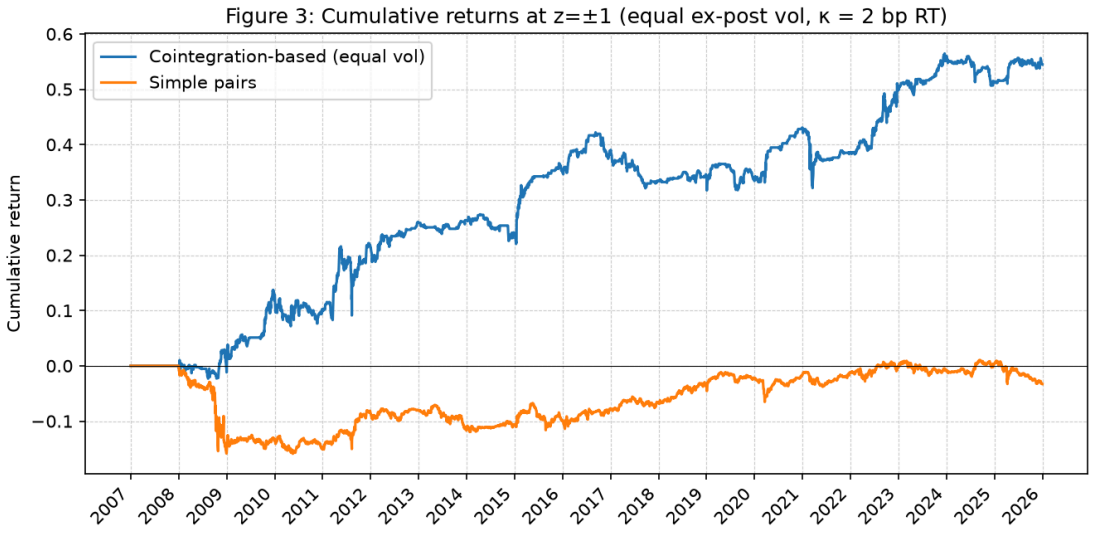


Figure 3: Cumulative returns at $z^\star = \pm 1$ (equal ex-post vol; $\kappa = 2$ bp; visuals only).

Unlevered and after costs, cointegration earns 0.49% annualized with an invested fraction of 8.2% of pair-days and 68% active days: most of the 1/21 book is cash on a typical day. Volatility and drawdown are correspondingly far lower than the always-trade rule, whose 2 bp haircut turns its already thin $z^\star = 1$ edge slightly negative. The equal-vol equity curve rises more steadily than the simple path, which suffers deeper early-sample drawdowns (notably around 2008 and 2011–12).

Widening the entry band to ± 2 (Table 3, Figure 4) reduces trading intensity and absolute returns for the cointegration book, but the filter retains a clear risk-adjusted edge after costs.

Metric	**Cointegration**	**Simple**
Annualized return (%)	0.25	0.18
Annualized volatility (%)	0.60	2.85
Sharpe ratio	0.41	0.06
Sortino ratio	0.34	0.06
Calmar ratio	0.13	0.01
Maximum drawdown (%)	−1.93	−12.51
Invested fraction (%)	2.97	19.25
Active days (%)	31.55	78.70

Table 3: Unlevered performance at $z^\star = \pm 2$ after $\kappa = 2$ bp RT costs. Occupancy from lagged signals (independent of κ).

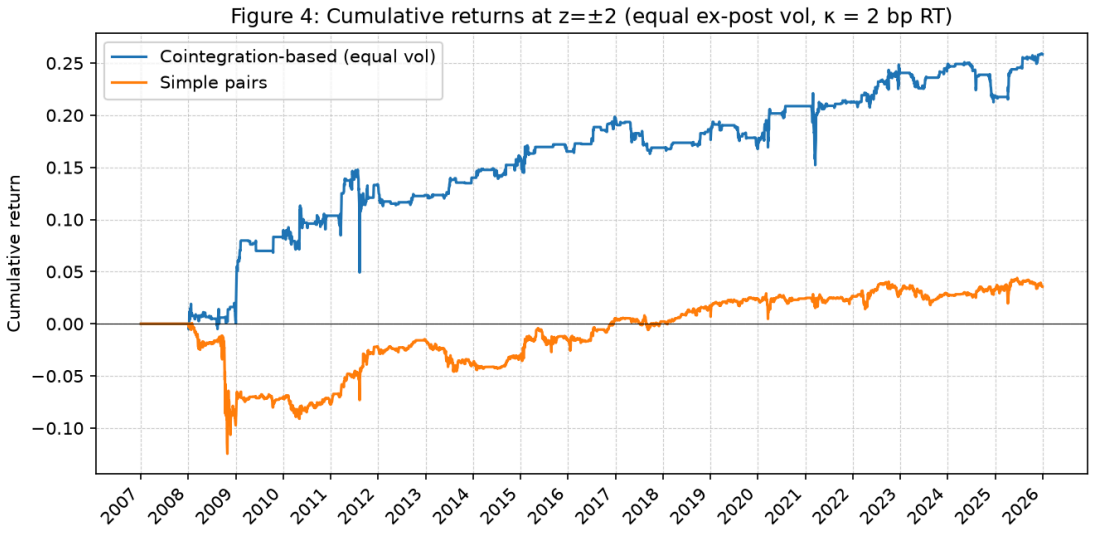


Figure 4: Cumulative returns at $z^\star = \pm 2$ (equal ex-post vol; $\kappa = 2$ bp; visuals only).

At ± 3 (Table 4, Figure 5) both books trade less often; cointegration still leads on Sharpe and related ratios after costs, with much smaller unlevered drawdowns.

Metric	**Cointegration**	**Simple**
Annualized return (%)	0.18	0.21
Annualized volatility (%)	0.38	2.06
Sharpe ratio	0.48	0.10
Sortino ratio	0.25	0.07
Calmar ratio	0.21	0.02
Maximum drawdown (%)	−0.86	−9.27
Invested fraction (%)	0.95	5.72
Active days (%)	11.10	38.15

Table 4: Unlevered performance at $z^\star = \pm 3$ after $\kappa = 2$ bp RT costs. Occupancy from lagged signals (independent of κ).

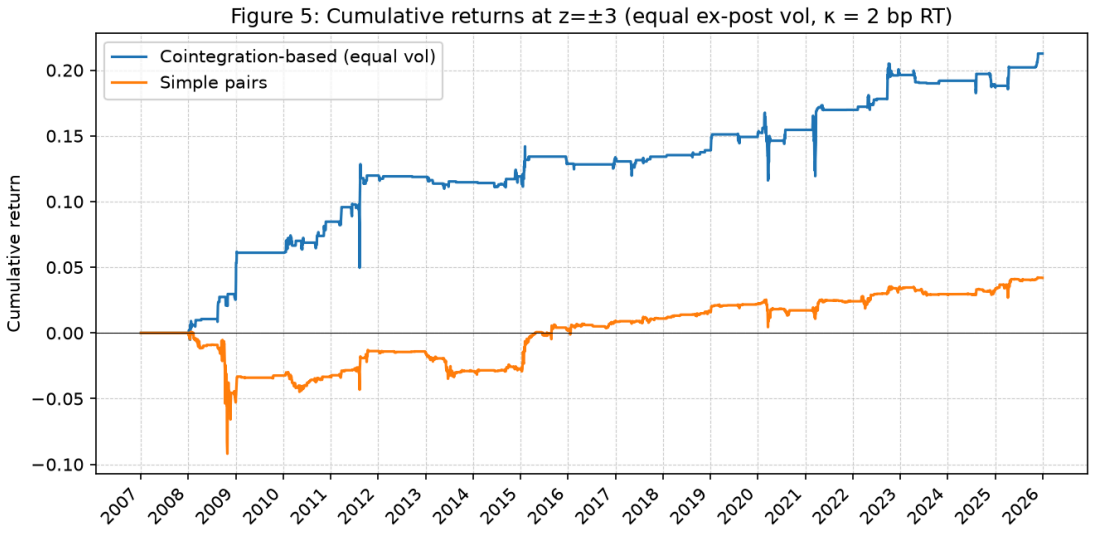


Figure 5: Cumulative returns at $z^\star = \pm 3$ (equal ex-post vol; $\kappa = 2$ bp; visuals only).

Table 5 and Table 7 summarize the threshold comparative at $\kappa = 2$. Table 8 shows that raising κ compresses both Sharpes, but the EG advantage remains positive through 5 bp at every threshold; at ± 1 the simple book is already near zero without costs and negative once $\kappa \geq 1$.

z-score	Cointegration Sharpe	Simple Sharpe
± 1	0.55	-0.04
± 2	0.41	0.06
± 3	0.48	0.10

Table 5: Unlevered Sharpe ratios across thresholds after $\kappa = 2$ bp RT costs.

Table 6 reports dependence-robust inference on those portfolio gaps after $\kappa = 2$ bp. At the headline threshold ± 1, ΔSharpe is $+0.59$ ($p = 0.014$; 95% studentized CI $[+0.12, +1.07]$) and ΔSortino is $+0.68$ ($p = 0.018$; 95% percentile CI $[+0.12, +1.28]$). At ± 2 and ± 3 the cointegration book still leads in point estimates, but the gaps are not statistically separable at conventional levels. The ± 1 conclusion is unchanged across the cost grid $\kappa \in \{0, 1, 2, 5\}$ and across block lengths of 5, 21, and 63 days; a HAC delta-method Sharpe test (Jobson & Korkie, 1981; Memmel, 2003) agrees with the bootstrap (replication package).

z	Statistic	Δ	p	95% CI	Method
± 1	Sharpe	$+0.59$	0.014	$[+0.12, +1.07]$	Studentized
	Sortino	$+0.68$	0.018	$[+0.12, +1.28]$	Percentile
± 2	Sharpe	$+0.35$	0.145	$[-0.14, +0.83]$	Studentized
	Sortino	$+0.27$	0.270	$[-0.22, +0.73]$	Percentile
± 3	Sharpe	$+0.37$	0.089	$[-0.07, +0.82]$	Studentized
	Sortino	$+0.18$	0.332	$[-0.21, +0.46]$	Percentile

Table 6: Portfolio ΔSharpe and ΔSortino (EG $-$ simple) on the unlevered 1/21 book after $\kappa = 2$ bp. Paired circular block bootstrap ($B = 10\{,\}000$; block length 21 days; Newey–West lag 9 for studentized Sharpe). Sortino uses the paper's downside definition (std of strictly negative daily returns) with a percentile interval. ± 1 is the pre-committed headline design.

z-score	Cointegration MDD (%)	Simple MDD (%)
± 1	-2.48	-15.89
± 2	-1.93	-12.51
± 3	-0.86	-9.27

Table 7: Unlevered maximum drawdowns by threshold after $\kappa = 2$ bp RT costs.

z-score	κ (bp)	EG Sharpe	Simple Sharpe
± 1	0	0.62	0.01
	1	0.58	−0.02
	2	0.55	−0.04
	5	0.45	−0.12
± 2	0	0.45	0.11
	1	0.43	0.08
	2	0.41	0.06
	5	0.35	−0.00
± 3	0	0.50	0.13
	1	0.49	0.11
	2	0.48	0.10
	5	0.43	0.06

Table 8: Unlevered Sharpe sensitivity to round-trip cost κ (bp of pair notional).

Table 9 reports capital deployment. At ± 1, the cointegration book has an invested fraction of 8.2% of pair-days versus 51% for always-trade; the Engle–Granger gate itself is on only 18.5% of pair-days. Occupancy falls further at wider thresholds. These figures explain the tiny unlevered volatilities.

z	Strategy	Invested (%)	Active days (%)	Live | active	Gate (%)
± 1	Cointegration	8.24	68.33	2.53	18.55
	Simple	51.27	94.58	11.38	94.72
± 2	Cointegration	2.97	31.55	1.97	18.55
	Simple	19.25	78.70	5.14	94.72
± 3	Cointegration	0.95	11.10	1.80	18.55
	Simple	5.72	38.15	3.15	94.72

Table 9: Portfolio occupancy from lagged signals (independent of κ). Invested fraction is the mean across days of live pairs /21; active days are days with at least one live pair; live | active is the mean number of live pairs on those days; gate occupancy is the share of pair-days with a non-NaN z-score (tradability-filter pass). Occupancy explains why unlevered volatility is small.

Table 10 applies that rule after $\kappa = 2$ bp at $z^\star \in \{1, 2, 3\}$. Both strategies use the same 3% train-window sizing, so book volatility follows how many pairs are on. At ± 1 the cointegration Sharpe edge survives (0.47 vs 0.09) with 3.87% return at 8.29% book vol. Sortino is higher than Sharpe for the cointegration companion (0.79 vs 0.47)—and higher than the unlevered EG Sortino of 0.63—because Sharpe penalizes right-tail volatility; the simple book does not show that gap (0.11 vs 0.09). Its about 22% vol and −68% drawdown is the cost of sizing many live pairs the same way. At ± 2 and ± 3 cointegration still leads on Sharpe; simple has higher raw return because more pairs are live.

z	Strategy	Ann. ret. (%)	Ann. vol. (%)	Sharpe	Sortino	Calmar	MDD (%)
±1	Cointegration	3.87	8.29	0.47	0.79	0.20	−18.9
	Simple	1.91	21.97	0.09	0.11	0.03	−68.0
±2	Cointegration	1.65	4.15	0.40	0.35	0.17	−9.8
	Simple	2.21	14.42	0.15	0.17	0.06	−37.2
±3	Cointegration	1.24	3.12	0.40	0.22	0.12	−10.6
	Simple	2.17	9.81	0.22	0.18	0.08	−28.0

Table 10: Companion: each live pair sized to 3% train-window vol ($L = 0.03/\hat{\sigma}$; $\kappa = 2$ bp). Same rule on both books; book = sum of pair PnLs (not 1/21). The 3% cap is an arbitrary choice. Book vol follows how many pairs are live. Mean $L \approx 0.33$.

5 Conclusions and future research

This study asks whether an Engle–Granger–style tradability filter improves FX pairs trading among seven liquid USD crosses. On rolling 257/21 windows and 21 unordered pairs, unlevered cointegration-book returns are small (about 0.2–0.5% annualized after 2 bp) because occupancy is low: at ±1 the invested fraction is about 8% of pair-days. The filter still leads the always-trade benchmark on Sharpe and Sortino at $z^\star \in \{1, 2, 3\}$ in point estimates. A paired block-bootstrap test finds that those risk-adjusted gaps are statistically significant at the pre-committed headline threshold ±1, but not separable at ±2 or ±3. The ranking matches the frictionless case and remains positive through 5 bp. Sizing each live pair to 3% train-window volatility raises cointegration to about 4% annualized return at about 8% book vol. Wider thresholds further reduce unlevered volatility and drawdowns but shrink occupancy and the incremental Sharpe benefit of the filter.

The residual stage of the filter uses an ordinary ADF cutoff as a fixed tradability rule, not as a MacKinnon-sized claim that cointegration was detected at 5%. Tightening that cutoff to formal Engle–Granger critical values is a natural robustness check; so are a broader grid of window lengths and thresholds, parity-based filters (PPP or UIP residuals), overnight swap costs, and Johansen filters.

Data and code availability. Replication code, the frozen Yahoo-derived daily FX sample (`data/fx_prices.parquet`), and this manuscript are available at https://github.com/QuantLandi/fx-cointegration. Code, scripts, and data are released under the MIT License; Typst source, PDF, figures, and bibliography under `paper/` under CC BY 4.0. Pipeline instructions are in the repository README.

Conflict of interest. The authors declare no potential conflict of interest.

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